

SECOND SEMESTER (CBCSS—UG) DEGREE EXAMINATION, APRIL 2021

Mathematics

MAT 2C 02 MATHEMATICS---2

(2020 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. Prove that $\cosh^2 x - \sinh^2 x = 1$.
2. Find the Cartesian form of the polar equation $r = \frac{8}{1 - 2\cos\theta}$.
3. Find the slope of the line tangent to the graph of $r = 3\cos^2 2\theta$ at $\theta = \pi/6$.
4. Evaluate $\int \sinh^2 x dx$.
5. Show that $\lim_{n \rightarrow \infty} \frac{2n}{n^2 + 1} = 0$.
6. Test the convergence of the series $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{4}} - \frac{1}{\sqrt{8}} + \frac{1}{\sqrt{16}} \dots$
7. Compute $\|\cos x\|$ in $C[0, 2\pi]$.
8. Examine whether the set of vectors $u_1 = \langle 1, 2, 3 \rangle, u_2 = \langle 2, 4, 3 \rangle$, and $u_3 = \langle 3, 2, 1 \rangle$ is linearly independent or not.
9. Find the eigenvalues of the matrix $A = \begin{bmatrix} 3 & 4 \\ -1 & 7 \end{bmatrix}$.
10. Find the determinant of the matrix $C = \begin{bmatrix} -1 & 2 & 9 \\ 2 & -4 & -18 \\ 5 & 7 & 27 \end{bmatrix}$.

Turn over

11. Show that $A = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ is an orthogonal matrix.

12. Find the eigen values of the matrix $A = \begin{bmatrix} 10 & 3 \\ 4 & 6 \end{bmatrix}$.

(8 × 3 = 24 marks)

Section B

Answer at least five questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Find the length of the curve $y = \frac{4\sqrt{2}}{3}x^{3/2} - 1, 0 \leq x \leq 1$.

14. Find the equation of the tangent line when $t = 1$ for the curve $x = t^4 + 2\sqrt{t}, y = \sin(t\pi)$.

15. Find the length of the perimeter of the cardioid $r = a(1 - \cos\theta)$.

16. Use the Trapezoidal rule with $n = 4$ to estimate $\int_1^2 x^2 dx$. Compare the estimate with the exact value of the integral.

17. Using Maclaurin's series expand $\tan^{-1} x$. Hence deduce the Gregory series $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

18. Show that the set $B = \{(1, 2, 1), (2, 1, 0), (1, -1, 2)\}$ is a basis for $\mathbb{R}^3$.

19. Find the inverse of the matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 0 \\ 3 & -1 & 4 \end{bmatrix}$.

(5 × 5 = 25 marks)

Section C

Answer any **one** question.
The question carries 11 marks.

20. (a) Evaluate $\int_1^{\infty} \frac{\ln x}{x^2} dx$, if it exists.

(b) Find the area of the region shared by the cardioids $r = 2(1 + \cos \theta)$ and $r = 2(1 - \cos \theta)$.

21. (a) Solve :

$$x_1 + x_2 + x_3 + x_4 = 0$$

$$x_1 + 3x_2 + 2x_3 + 4x_4 = 0$$

$$2x_1 + x_3 - x_4 = 0.$$

(b) Find the eigen values of the matrix $A = \begin{bmatrix} 1 & -6 \\ 2 & 2 \end{bmatrix}$.

(1 × 11 = 11 marks)

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APRIL 2021

Mathematics

MAT 2C 02—MATHEMATICS—2

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

Answer at least **eight** questions.

Each question carries 3 marks

All questions can be attended.

Overall Ceiling 24.

1. Compute the derivative of $\sqrt{x}$ using inverse function rule. Evaluate the derivative at $x = 2$.
2. Convert the relation $r = 1 + 2 \cos \theta$ to Cartesian co-ordinates.
3. Compute $\int \cosh^2 x \, dx$.
4. Find $\frac{d}{dx} \cosh^{-1} \sqrt{x^2 + 1}$, $x \neq 0$.
5. Find $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$.
6. Show that $\int_0^{\infty} \frac{\sin x}{(1+x^2)} dx$ converges.
7. A bouncing ball loses half of its energy on each bounce. The height reached on each bounce is proportional to the energy. Suppose that the ball is dropped vertically from a height of one meter. How far does it travel ?

Turn over

8. State Ratio comparison test and show that $\sum_{i=1}^{\infty} \frac{2}{4+i}$ diverges.
9. Prove that the vectors $w_1 = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$, $w_2 = \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$ and $w_3 = \left(0, \frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ are orthonormal vectors.
10. Define basis of a vector space. Give a basis for vector space P_n of all polynomial of degree less than or equal to n .
11. Find the inverse of $A = \begin{pmatrix} 1 & 8 \\ 2 & 10 \end{pmatrix}$.
12. Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}$.

(8 × 3 = 24 marks)

Section B

Answer at least five questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Let $f(x) = x^2 + 2x + 3$. Restrict f to a suitable interval so that it has an inverse. Find the inverse function and sketch its graph.
14. Find the length of the graph of $f(x) = (x-1)^{3/2} + 2$ on $[0, 2]$.
15. State root test and test the convergence for the series $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$.
16. For which x does the series $\sum_{n=0}^{\infty} \frac{4^n}{\sqrt{2n+5}} (x+5)^n$ converge.

17. Let $u_1 = (1, 1, 1)$, $u_2 = (1, 2, 2)$ and $u_3 = (1, 1, 0)$ be basis of $\mathbb{R}^3$. Using Gram Schmidt process find an orthonormal basis of $\mathbb{R}^3$.

18. Compute A^m for $A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$.

19. Identify the conic whose equation is $2x^2 + 4xy - y^2 = 1$.

(5 × 5 = 25 marks)

Section C

Answer any one question.

The question carries 11 marks.

20. (a) Find the area of the surface obtained by revolving the graph $y = x^2$ about the y -axis for $1 \leq x \leq 2$.
- (b) Determine whether the set of vectors $u_1 = (1, 2, 3)$, $u_2 = (1, 0, 1)$ and $u_3 = (1, -1, 5)$ is linearly dependent or linearly independent.
21. (a) Find the terms through cubic order in the Taylor series for $\frac{1}{1+x^2}$ at $x_0 = 1$.

(b) Find an LU factorization of $A = \begin{pmatrix} -1 & 2 & -4 \\ 2 & -5 & 10 \\ 3 & 1 & 6 \end{pmatrix}$.

(1 × 11 = 11 marks)